Heat from Bit: The Thermodynamic Price of Objective Collapse

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Abstract

We propose the hypothesis that any mechanism of *objective* collapse must pay a thermodynamic price of $k \ln 2$ per destroyed bit of coherence *in the pointer basis of the act*—as heat $\Delta Q = kT \ln 2$ in an environment at temperature T. The tariff is evaluated on a state conditioned only on a physically delivered local pointer record. We prove that a nontrivial nonnegative tariff vanishing on pointer-diagonal states cannot be an affine functional of one local density matrix. Test: calorimetry of heating upon absorption of a heralded single-photon qubit—$|+\rangle$ versus the control $I/2$.

Keywords: objective collapse; pointer state; quantum Darwinism; Landauer's principle; heat from bit; entropy collection

1 Introduction

The idea of *objective* collapse—making state reduction a physical process rather than an epistemic postulate of measurement—forms a distinct research program alongside standard quantum mechanics, where the apparent collapse is explained by unitary decoherence and correlations with the environment. The early line of Karolyházy [1] linked the demise of superpositions to spacetime uncertainty; GRW and CSL [2, 3] introduced spontaneous localization; Diosi–Penrose models [4, 5]—gravitational instability of macroscopic superpositions; in other models: crossing a scrambling threshold [6], branch-dependent dephasing of proper time [7]. Review and status—Bassi et al. [8]: the direction is actively developed and experimentally constrained, but remains marginal relative to orthodox quantum theory; decisive confirmation or final refutation is still lacking. The present work does not choose a criterion for "when," but asks about the thermodynamic price of any such act.

Objective collapse introduces a fundamentally irreversible element of dynamics, distinguishing the preferred direction of time from reversible unitary evolution. However, such irreversibility by itself is not equivalent to violation of continuous time-translation symmetry, with which Noether's theorem associates energy conservation. Therefore the energetic footprint of collapse depends on the specific non-unitary dynamics; the present work proposes a hypothesis about its thermodynamic nature (§ 5).

Landauer's principle [9] links the erasure of *classical* information to minimum dissipation $Q \geq kT \ln 2$. The price sits on logically irreversible compression of phase volume (many→one)—a consequence of the second law [10, 11]. Under decoherence the environment selects a *pointer basis*—a set of robust outcomes $|i\rangle$, proliferated in the environment (einselection; quantum Darwinism [12]). The pointer basis defines the set of *classically admissible* (incoherent) states—diagonal mixtures over $|i\rangle$. The *relative entropy of coherence* [13] $C_r(\rho) = \min_{\sigma \in \mathcal{I}} S(\rho \| \sigma)$ measures how far ρ falls short of such a mixture; for the pointer basis the minimum is attained at ρ_{diag} and $C_r(\rho) = S(\rho_{\text{diag}}) - S(\rho)$. For pure $|+\rangle_{\text{ptr}}$ $S(\rho) = 0$, but $C_r = k \ln 2$—full which-path coherence; for $I/2_{\text{ptr}}$ $S(\rho) = k \ln 2$, but $C_r = 0$—the same pointer statistics without phase,

nothing to erase. Under *unitary* decoherence $|+\rangle \to I/2$ $S(\rho_S)$ of the subsystem grows to $k \ln 2$, but the entropy of the whole is zero: the phase goes into correlations and is recoverable in principle (Loschmidt echo [14, 15]). Under *objective* collapse the pointer outcome is fixed irreversibly, $H = S_{\rm coh} = C_r$ in the pointer basis—a thermodynamically active quantity; according to the hypothesis of the present work its jump materializes as heat. Hormoz [16] shows the necessity of an entropic price for collapse in a ratchet model; Heat from Bit quantifies the price for objective fixation of a pointer outcome without an explicit demon.

Wheeler formulated "It from Bit" [17]: physical reality ("it") arises from acts of distinguishing pointer outcomes. *Heat from Bit*—the same parallel in thermodynamics: $C_r(\rho)$ in the pointer basis is the "bit"; its irreversible jump is the "it," materializing as heat.

We propose an experimental test (§ 6): calorimetry of a *heralded* single-photon qubit in a Mach–Zehnder (MZ) interferometer (heralded MZ)—$|+\rangle_{\rm ptr}$ yields $kT \ln 2$ per act, the classical mixture $I/2_{\rm ptr}$—none.

2 Intuition

The idea of Heat from Bit grew out of a simple thought experiment in which *one* photon is absorbed by an anisotropic absorber in the polarization basis that defines the pointer basis of the act. The basis is $\{|0\rangle, |1\rangle\}$ of linear polarization ($|0\rangle$—horizontal, $|1\rangle$—vertical): which of these outcomes will be irreversibly fixed upon absorption—that is, upon detection?

Case A (definite polarization). A single-photon state $|0\rangle$—an eigenstate of the pointer basis, aligned with the polarizer axis. Coherence $S_{\rm coh} = 0$. The photon is absorbed, the absorber receives energy E_γ—ordinary first-law heating; the alternative pointer outcome $|1\rangle$ is absent in the pre-act state, nothing to erase. Nothing changes upon an act of objective collapse.

Case B (superposition of pointer outcomes). The same single-photon protocol, but the photon is prepared in

$$|+\rangle_{\rm ptr} = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \tag{1}$$

in the polarizer's pointer basis: $S_{\rm coh} = \ln 2$. Absorption transfers the same E_γ; the difference lies in the pre-act state, not in the photon energy.

Step 1 (absorption). Regardless of preparation, the act of absorption transfers E_γ to the absorber; there is as yet no objective collapse at this stage.

Step 2 (decoherence). Under *unitary* evolution, coherence in the pointer basis $S_{\rm coh}$ goes into the internal degrees of freedom of the absorber and environment—into entanglement, recoverable in principle (echo, reversible cycle). The environment already interacts with the photon and carries information about the pointer outcome (here—polarization) into the *internal memory* of the surroundings. Entanglement of the absorber and environment with $|0\rangle$ and $|1\rangle$ *is* that memory—the alternative is not destroyed but encoded in correlations; under unitary evolution it is preserved, there is no erasure, global entropy does not grow.

Step 3 (objective collapse). Objective collapse can be understood as *clearing* internal memory: degrees of freedom that carried coherence and entanglement disappear irreversibly—a register-erasure procedure, not a reversible transfer of phase. The pointer outcome is fixed; there is no longer an "internal" quantum state storing the alternative $|0\rangle$ vs $|1\rangle$, $S_{\rm coh}$ vanishes. This is an act of erasing one bit of pointer outcome: from the superposition $|+\rangle$ irreversibly one outcome $|0\rangle$ or $|1\rangle$ remains. We propose the hypothesis that the fundamental destruction of this information obeys a thermodynamic constraint analogous to Landauer's principle [9]: the environment of the act—the reservoir where pointer modes and their correlations with the absorber lived—must receive additional heat

$$\Delta Q = kT \ln 2 \tag{2}$$

for each objectively fixed single-photon act ($|+\rangle_{\rm ptr} \to |0\rangle$ or $|1\rangle$), above ordinary heating by E_γ. The excess of case B over A at equal E_γ—the prediction of Heat from Bit.

Case C (classical mixture). The same protocol, but the photon is prepared as $I/2$ in the pointer basis:

$$I/2 = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2} \tag{3}$$

—a diagonal mixture of $|0\rangle$ and $|1\rangle$ without phase correlations: $S_{\rm coh} = 0$. This is not a superposition of pointer outcomes but a statistical record: the outcome is already "measured" at the level of the pre-act state, only classical uncertainty remains as to *which* of them was realized. In the act of absorption the environment does not *erase* an alternative bit but merely *copies* an already existing classical record; copying classical information is energetically free. There is neither decoherence nor objective collapse in the act.

3 Pointer basis and coherence entropy

Upon interaction with the environment the system decoheres into a *pointer basis* $\{|i\rangle\}$—robust outcomes proliferated in the environment (einselection; quantum Darwinism [12]). A classical record is a specific pointer outcome $|i\rangle$. $|+\rangle$ and $I/2$ coincide in $\rho_{\rm ptr}$ and $\langle\hat{H}\rangle$; a bolometer sees only the diagonal in the pointer basis. The tariff is by coherence in the pointer basis *of the act*, not by $S(\rho)$ abstractly.

Coherence in the pointer basis. In the resource theory of coherence [13, 18]:

$$S_{\rm coh}(\rho) = S(\rho_{\rm ptr}) - S(\rho), \tag{4}$$

where $\rho_{\rm ptr}$ is the diagonal of ρ in the pointer basis of the act. For $|+\rangle$ $S_{\rm coh} = \ln 2$, for $I/2$—zero; this is the *relative entropy of coherence* $C_r(\rho)$ [13, 18] relative to the same basis. Kammerlander and Anders [19] show that the thermodynamics of *projection* into a basis (dephasing, nonselective measurement) is nontrivial only when there is coherence in that basis; for diagonal mixtures $\Delta S = 0$. In an optimal process for $|+\rangle$ from such projection one can extract work $\langle W\rangle = kT\ln 2$. Heat from Bit transfers this distinguishability to the pointer basis of the act and postulates dissipation linked to $S_{\rm coh}$ upon objective fixation of the outcome.

Tariff entropy H. Let H denote the *tariff entropy* of the pre-act state in the pointer basis *of the act*; $S(\rho)$, $S_{\rm coh}$, and H are in *nats*:

$$H(\rho) = S_{\rm coh}(\rho). \tag{5}$$

H is neither the full $S(\rho)$ nor the classical $-\sum_i p_i \ln p_i$, but precisely the coherence in the pointer basis destroyed upon irreversible fixation of the outcome; it sets the thermodynamic price. Diagonal $\rho_{\rm ptr}$ ($I/2$): $H = 0$; pure superposition $|+\rangle_{\rm ptr}$: $H = \ln 2$.

Number of bits:

$$N_{\rm bit}(\rho) = H(\rho)/\ln 2. \tag{6}$$

Coherent states of the field ($\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$) are eigenstates of the operator $\hat{a}$ [20]; they define the classical amplitude of a mode and a CW laser beam. The term "coherent" here does not mean $S_{\rm coh}$ in the pointer basis: in $|\alpha\rangle$ there is no superposition of pointer outcomes, only Poisson statistics, $H = 0$; a "photon" of a coherent beam has no separate pre-act ρ. "Angle 45°" of a CW laser does not reproduce case B from § 2:

$$|\alpha\rangle_{45^\circ} = |\alpha/\sqrt{2}\rangle_0\, |{\rm i}\alpha/\sqrt{2}\rangle_1, \tag{7}$$

not a flux of $|+\rangle_{\rm ptr}$. For a pure pre-act $\rho = |\psi\rangle\langle\psi|$, $|\psi\rangle = \sum_i c_i|i\rangle$, we have $p_i = |c_i|^2$ and $H = -\sum_i p_i \ln p_i$ (then $S(\rho) = 0$). In the general case $H = S(\rho_{\rm ptr}) - S(\rho)$; for mixed $I/2$ this is $H = 0$, although $-\sum_i p_i \ln p_i = \ln 2$.

4 Thermodynamics of objective collapse

The tariff H (§ 3) is a definition ($H = S_{\text{coh}}$ in the pointer basis of the act), not an assumption of the model.

Under *unitary* dephasing $S(\rho_{\text{ptr}})$ does not change, the growth of $S(\rho)$ of the subsystem is the decrease of S_{coh} (§ 3); an isolated subsystem under unitary $\pi/2$ does not change $S(\rho)$ (pure remains pure), and a reversible cycle $\pi/2 \to -\pi/2$ restores the state; tariff H is not activated. Heat does *not* follow from the growth of $S(\rho)$ of the subsystem and does not accompany unitary "encryption" of the pointer-basis phase into correlations with the environment.

Postulate (Heat from Bit). Testable hypothesis of the work (§ 6): under *objective* collapse with $H > 0$ the jump of the tariff is identified with thermodynamic entropy of the reservoir $S_{\text{th}} = kH$ and heat $\Delta Q = T\Delta S_{\text{th}} = kT\Delta H$ in the environment at temperature T; the pointer outcome is fixed irreversibly. Average over acts

$$\langle Q\rangle = kT\,H = kT\,C_r(\rho). \tag{8}$$

For a given H the distribution of Q_i does *not* follow from this (for example, $Q_i \equiv kTH$ gives the same $\langle Q\rangle$). For a *pure* pre-act $|\psi\rangle = \sum_i c_i|i\rangle$ ($S(\rho) = 0$, $p_i = |c_i|^2$) the model *additionally* adopts a per-trajectory tariff

$$\boxed{Q_i = kT\ln(1/p_i)} \tag{9}$$

—informational "surprise" of outcome i (consistent with the average via $H = -\sum p_i \ln p_i$): a rare outcome "erases" more alternative probability mass. Then $\Delta S_{\text{th},i} = k\ln(1/p_i)$,

$$\langle Q\rangle = \sum_i p_i\,Q_i = kTH, \qquad \mathrm{Var}(Q) = (kT)^2\,\mathrm{Var}(\ln p_i). \tag{10}$$

For a mixed state diagonal in the pointer basis ($H = 0$, e.g. $I/2$) the per-act tariff is not activated: $\langle Q\rangle = 0$; a partially coherent mixed state generally has $H > 0$. At equal $p_i = 1/n$ all outcomes of a pure state are equivalent: $Q_i = kT\ln n$; for binary $|+\rangle$ ($p_i = \frac{1}{2}$) this is $\Delta Q = kT\ln 2$, $\Delta H = \ln 2$ and $\Delta S_{\text{th}} = k\ln 2$ ($\Delta N_{\text{bit}} = 1$). Example $p_0 = 0{,}9$, $p_1 = 0{,}1$: an act with outcome 0 releases $kT\ln(10/9) \approx 0{,}11\,kT$, with outcome 1—$kT\ln 10 \approx 2{,}3\,kT$; the average $\approx 0{,}33\,kT = kT\,h_2(0{,}9)$.

The per-outcome tariff is consistent with Born's rule and stochastic thermodynamics [11]. At equal amplitudes ($|+\rangle$) Q_i are equal, no fluctuations; the difference between versions arises only for asymmetric superposition and per-act calorimetry (currently unattainable; § 6).

Proliferation of an already existing record (CNOT, redundancy in the sense of quantum Darwinism [12]) does not create a new bit; unitary entanglement (GHZ, cat states) does not multiply the tariff. Unitary preparation of $|+\rangle_{\text{ptr}}$ ($\pi/2$, which-path encoding $|x_0\rangle + |x_1\rangle$, etc.)—reversible operations; no price.

On a series of objective acts (average over outcomes):

$$dH = \ln 2\,dN_{\text{bit}}, \qquad \langle \delta Q\rangle = kT\,dH = \mu_{\text{bit}}\,dN_{\text{bit}}, \qquad \mu_{\text{bit}} = kT\ln 2. \tag{11}$$

For a pure pre-act state one act with outcome i: $\Delta Q_i = kT\ln(1/p_i)$; $\langle \Delta Q\rangle = kT\,\Delta H = kT\ln 2\,\Delta N_{\text{bit}}$. The payment is not $\langle \hat{H}\rangle$ of the Hamiltonian but the thermodynamic price of fixing a pointer outcome; δQ is dissipated in the reservoir, on average—at tariff μ_{bit}.

4.1 The state on which the tariff is evaluated

Equation (8) is incomplete without a rule specifying which local state enters C_r. We distinguish three possibilities. (i) *Unconditioned local rule*: $Q = kTC_r(\rho_S)$ for the reduced state of the act system. It is causal and cannot distinguish ensemble decompositions of the same ρ_S, but an averaged phase scrambler becomes the same $I/2$ as the classical control. (ii) *Locally*

conditioned rule: if an objective classical pointer register R carrying record r has physically entered the causal state of the act region, use the conditional state $\rho_{S|r}$,

$$\langle Q\rangle_R = kT \sum_r p_r\, C_r(\rho_{S|r}). \tag{12}$$

(iii) *Globally conditioned rule*: a remote ontic branch is selected even without a locally delivered record; this stronger version can distinguish remotely prepared decompositions of one ρ_S.

This work adopts rule (ii). A "record" is a physical local register causally delivered to the act, not an observer's knowledge or information merely existing somewhere in its past light cone. If R itself remains in superposition or is only entangled with S, no classical index r is introduced and no arbitrary ensemble decomposition of ρ_S may be selected. Equation (12) therefore requires an external criterion for when R has become an objective pointer register; Heat from Bit specifies the price of such an act, not a universal criterion for "when".

Theorem for a linear tariff. Let a single-act sector be finite-dimensional, let $\{|i\rangle\}$ be a complete pointer basis, and let the mean tariff be affine in the local density matrix. Then $Q(\rho) = \mathrm{Tr}(X\rho)$ for a Hermitian X (a constant is absorbed into the identity component). If

$$Q(|i\rangle\langle i|) = 0 \quad \forall i, \qquad Q(\rho) \geq 0 \quad \forall \rho, \tag{13}$$

then $X = 0$ and $Q \equiv 0$. Indeed, the second condition gives $X \geq 0$, the first gives $X_{ii} = 0$, and positivity implies $|X_{ij}|^2 \leq X_{ii}X_{jj} = 0$. Thus a *nontrivial* positive tariff that vanishes on every pointer-diagonal state cannot be an affine functional of one local ρ. This is a triviality theorem for the linear branch, not an equivalence "nonlinearity $\Leftrightarrow$ signaling".

Affineness of a local observable is sufficient for different decompositions of one ρ_S not to signal through that channel; the converse is false. In particular, nonlinear $kTC_r(\rho_S)$ also does not signal: for one half of a Bell pair $\rho_B = I/2$ in every remote basis and $C_r(I/2) = 0$. A signal arises only after the additional global conditioning (iii), not from nonlinearity alone [21, 22, 23, 24].

5 Origin of heat energy

In concrete objective-collapse models the non-unitary dynamics itself may specify energy released upon localization: in CSL and GRW [2, 3, 8] the collapse operator raises the mean energy of the system. Then tariff heat $\delta Q = kT\Delta H$ (§ 4) could in principle come from that reservoir. In other models such a reservoir may not be explicit or may be negligible.

We consider the duration of the collapse act τ_{col}—non-unitary compression of the superposition to a single pointer outcome. Under decoherence the system entangles with environmental degrees of freedom of the act—"receptors" of pointer information; the reservoir at the absorber temperature T is populated with modes on the scale kT, and the characteristic time of thermal oscillations is $\tau_{\mathrm{th}} \equiv \hbar/kT$. In many objective-collapse theories the moment of localization depends on the number of environmental particles decohered with the system or on the effective mass in the superposition [2, 8, 12, 6, 7]. This mass grows as new "receptors" join over a time of order τ_{th}. The value τ_{col} is set by the particular collapse model; however, heuristically, when the criterion grows with the number of receptors, non-unitary compression must complete within $\tau_{\mathrm{col}} \lesssim \tau_{\mathrm{th}}$—before one more environmental degree of freedom joins the superposition and the criterion shifts.

In an act of duration τ_{col} energy is defined no better than $\Delta E \sim \hbar/(2\tau_{\mathrm{col}})$ (the Mandelstam–Tamm relation [25] bounds the *rate* of change of an observable); for the full payment $kT\ln 2$ to fit in this window, we link τ_{col} to the tariff payment $\delta Q = kT\ln 2$:

$$\tau_{\mathrm{col}} \lesssim \frac{\hbar}{2kT\ln 2} \approx \frac{\tau_{\mathrm{th}}}{2\ln 2}, \tag{14}$$

A single act is not microscopically auditable; over a series of N acts one observes the total excess $\Delta Q_{\text{tot}} = N\,kT\Delta H$ (§ 6). Under *unitary* evolution the mean Hamiltonian is conserved, but objective collapse is non-unitary, therefore the mechanism forbidding local energy imbalance is switched off at the moment of pointer-outcome fixation. Thus, in the absence of an explicit reservoir in the collapse model, the energy for tariff heat δQ may be drawn from the energy uncertainty of the act duration and dissipated in the thermal reservoir.

6 Experimental proposal

The proposed test is differential: the postulated term *adds* heat to ordinary heating ($Q_{\text{out}} = Q_{\text{std}} + \mu_{\text{bit}}\,\Delta N_{\text{bit}}$ when $\Delta N_{\text{bit}} > 0$), therefore a systematic excess over a control with $H = 0$ is measured. Here Q_{std} is heat without tariff (absorption W_{in}, ordinary Q_{dissip}); ΔN_{bit} is the number of pointer bits objectively erased in acts (§ 3): one act from $|+\rangle_{\text{ptr}}$ gives $\Delta N_{\text{bit}} = 1$, from $I/2_{\text{ptr}}$—zero. Heralded MZ—$\sim 10^{10}$ at $R = 10^8$ s^{-1} and $\tau \sim 10^2$ s; all test $\langle Q\rangle = kTH$ (§ 4), not distinguishing fluctuations of Q_i by outcome. A test correlating Q with outcome requires asymmetric superposition and per-act calorimetry and is *currently* unattainable: $\Delta Q_i \sim 10^{-2}$ eV per herald at $T = 300$ K is comparable to single-event bolometer noise, while heralded MZ targets $\langle Q\rangle$ via lock-in and accumulation of $\sim 10^{10}$ acts.

An ordinary CW laser after 50/50 is a product of coherent modes $|\alpha/\sqrt{2}\rangle_L|\mathrm{i}\alpha/\sqrt{2}\rangle_R$, not a flux of $|+\rangle_{\text{ptr}}$ qubits; the same for polarization $|\alpha\rangle_{45^\circ}$ (§ 3). Here—a pure flux of qubits: SPDC + SNSPD herald $\to |1\rangle_{\text{Fock}} \to$ Mach–Zehnder $\to$ two pointer zones of absorption.

Pointer basis and states. The pointer basis of the act is *localization of absorption*: $|x_0\rangle$, $|x_1\rangle$—two separated zones on one micro-island (left and right arm of the M–Z). Single-photon path state after the first M–Z splitter:

$$|\psi\rangle = \frac{|1,0\rangle + e^{\mathrm{i}\phi}|0,1\rangle}{\sqrt{2}} \quad \Leftrightarrow \quad |+\rangle_{\text{ptr}} :\ H = S_{\text{coh}} = C_r = \ln 2. \tag{15}$$

Preparation is unitary ($S(\rho) = 0$); the tariff is activated only upon objective collapse (§ 4), not upon splitting or phase rotation. Control—*classical* path preparation without coherence (not a phase scrambler):

$$\rho_{\text{mix}} = \tfrac{1}{2}|1,0\rangle\langle 1,0| + \tfrac{1}{2}|0,1\rangle\langle 0,1| \quad \Leftrightarrow \quad I/2_{\text{ptr}} :\ H = 0; \tag{16}$$

diagonals of ρ_{ptr} coincide with $|+\rangle_{\text{ptr}}$, but $C_r(\rho_{\text{mix}}) = 0$. The control is implemented *before* the interferometer: an optical switch driven by an objective local bit r (physically $|1,0\rangle$ or $|0,1\rangle$ per herald). Under the adopted locally conditioned rule (12), the conditional state is already a pointer state and $H = 0$.

Fast phase randomization is not equivalent to this control if ϕ is objectively recorded in a local modulator register that causally enters the preparation of the act: conditioned on that record the state remains pure $|\psi(\phi)\rangle$ and $H = \ln 2$. If the phase controller remains quantum, its record is erased before the act, or it does not physically enter the causal state of the absorber, conditioning is forbidden and the unconditioned $I/2$ with $H = 0$ is used. The distinction is therefore fixed by a physical preparation register, not by an observer choosing between equivalent decompositions of one ρ (§ 4.1).

Why which-path, not polarization. § 3 shows the same tariff on a single-photon $|+\rangle_{\text{ptr}}$ in the polarization basis; heralded MZ implements an equivalent $|+\rangle_{\text{ptr}}$ in the localization basis $|x_i\rangle$—more convenient for event-triggered calorimetry and control of $I/2_{\text{ptr}}$.

Setup.

- **Source**: SPDC in a monolithic or external resonator (cavity SPDC) or several parallel SPDC pairs on a common herald; $\lambda = 1550$ nm, SNSPD on idler; signal $|1\rangle_{\text{Fock}} \to$ MZ (50/50, ϕ fixed). Design herald rate $R = 10^8$ s^{-1} (by $\sim$ 1–2 orders of magnitude above typical waveguide SPDC via resonator and/or multiplexing).
- **Absorber**: suspended micro-island (SiN/graphene/Ge–InGaAs), two pointer zones on one island; $T = 300$ K. Thermal link to substrate $G \sim 1$ μW/K (not mW/K: otherwise the phonon limit NEP $= \sqrt{4k_BT^2G}$ makes lock-in infeasible). Quantum efficiency of the path "herald $\to$ absorption in pointer zone" $\eta = 0{,}85$ (SNSPD, AR coatings, focus on zone).
- **Synchronization**: herald pulse $\to$ bolometer integration window $\ll T_2^*$ of path; lock-in upon switching $|+\rangle_{\text{ptr}} \leftrightarrow I/2_{\text{ptr}}$ at $f_{\text{mod}} \sim 1$ kHz.

Signal. For *each* objectively fixed heralded act in channel $|+\rangle_{\text{ptr}}$: $\Delta Q_{\text{bit}} = kT\ln 2 \approx 1{,}8 \times 10^{-2}$ eV ($\sim 2{,}2\%$ of E_γ at $\lambda = 1550$ nm); in $I/2_{\text{ptr}}$—zero. At design $R = 10^8$ s^{-1} and $\eta = 0{,}85$ the average excess power

$$\Delta P = \eta\, R\, kT \ln 2 \approx 2{,}4 \times 10^{-13}\ \text{W} \quad (T = 300\ \text{K}). \tag{17}$$

The fraction $\Delta Q_{\text{bit}}/E_\gamma$ is constant; verification is by slope $\Delta P/R$ and event-triggered calorimetry. At $R = 10^8$ s^{-1} contamination from two-pair SPDC events ($g^{(2)}(0) > 0$) dilutes the differential $|+\rangle_{\text{ptr}} - I/2_{\text{ptr}}$; $g^{(2)}(0) \ll 1$ on herald or low duty cycle of the source is required.

Noise budget. Phonon limit of the island: NEP $= \sqrt{4k_BT^2G}$. At design $G \sim 1$ μW/K NEP $\approx 2{,}2 \times 10^{-12}$ W/$\sqrt{\text{Hz}}$. Lock-in integration time to SNR ~ 1 in power: $\tau \approx (\text{NEP}/\Delta P)^2 \sim 10^2$ s at the ΔP above.

Channels (equal R, equal η). Channel $|+\rangle_{\text{ptr}}$: MZ with fixed ϕ; $\langle Q\rangle = kT\ln 2$ per heralded act.

Channel $I/2_{\text{ptr}}$: same R; before MZ classical which-path (L/R switch); $\langle Q\rangle = 0$.

The differential $|+\rangle_{\text{ptr}} - I/2_{\text{ptr}}$ isolates the price for one pointer bit. At $R = 10^8$ s^{-1} over $\tau \sim 10^2$ s $\sim 10^{10}$ acts accumulate and total excess $\Delta Q_{\text{tot}} \sim 10^{-11}$ J.

What we record.

1. Event-triggered energy excess $\Delta Q_{\text{bit}} \approx kT\ln 2$ per herald in channel $|+\rangle_{\text{ptr}}$; zero in $I/2_{\text{ptr}}$.
2. Linearity of total ΔQ in R and T; slope $\Delta P/R \propto \eta\, kT\ln 2$.
3. Lock-in peak at f_{mod}; absence of peak at constant $I/2_{\text{ptr}}$.

Outcomes. Confirmation: stable differential $|+\rangle_{\text{ptr}} - I/2_{\text{ptr}}$ at $R = 10^8$ s^{-1}. Null at SNR $\gg 1$ $\Rightarrow$ bound on tariff $kT\ln 2$.

Causal delayed-record test. Equation (12) permits an additional test without superluminal communication. Alice measures one half of a Bell pair in the Z or X basis and sends Bob a classical message containing *both basis and outcome*. If Bob's act occurs before physical arrival of the record, only $\rho_B = I/2$ is locally available and the tariff is zero. After an equal-size record arrives, the conditional states are $|0\rangle, |1\rangle$ for Z ($H = 0$) or $|+\rangle, |-\rangle$ for X ($H = \ln 2$). The "act before/after record" differential tests dependence on causally available side information; ordinary heat of delivery and memory must cancel between messages of equal energy. The same test can be implemented with a locally correlated register, so it probes the conditioning rule rather than quantum nonlocality itself.

Transmon as an outlook. Dispersive readout of a transmon at millikelvin temperatures is attractive because substrate temperature and readout photon number can be varied independently. Its sensitivity cannot, however, be obtained by extrapolating $G \propto T^3$ from room temperature. The thermal-conductance quantum $G_Q = \pi^2 k_B^2 T/(3h)$ is the maximum contribution of a perfectly transmitting channel, not a universal lower bound; a real budget must include the device-specific G, heat capacity and bandwidth, χ, κ, n_{crit}, pulse shape, photon shot noise, and ordinary state-dependent heat. We therefore retain the transmon only as a direction for a separate calculation, without a claimed integration-time table [26, 27].

7 Discussion

The Heat from Bit model postulates a single thermodynamic price—for objective collapse ($Q_i = kT\ln(1/p_i)$ for pure coherent states, $\langle Q\rangle = kTH$; § 4); unitarity is free. The existence of such acts is taken as a premise (introduction); the specific criterion for "when" the article does not fix. A substantive part of the postulate is the state rule: the working version conditions only on a physically delivered objective local pointer register (§ 4.1). The trajectory tariff Q_i remains an independent additional postulate, not a consequence of the linear-tariff theorem. The primary experimental test is heralded MZ (§ 6).

Thermal sector and detailed balance. Thermal states admit a positive P representation as mixtures of coherent $|\alpha\rangle$ [20], but for a nonlinear tariff that mathematical decomposition cannot automatically be treated as the physically realized ensemble. Under the unconditioned local rule a thermal ρ is diagonal in the relevant pointer basis and gives $H = 0$. Under the adopted conditioned rule one must additionally specify whether an objective local source register exists and which conditional states $\rho_{S|r}$ it physically prepares. An arbitrary convenient P decomposition is not allowed; no conditioning is made on a quantum or remote register.

This is an open compatibility condition with detailed balance, not a proved exemption of equilibrium light. If each absorbed solar photon of characteristic energy 2 eV paid even $kT\ln 2 = 0.0179$ eV at 300 K, the addition would be about 0.9%: for absorbed flux of order 240 W/m^2 this is $\sim$ 2.1 W/m^2, already above the measured imbalance $\sim$ 0.9 W/m^2 [28]. Splitting objectification into $O(1)$ acts limits the maximum to one bit but does not restore a zero tariff. FIRAS [29], Earth's energy balance, and laboratory detailed balance therefore constrain any admissible thermal-register rule.

Spontaneous emission and tariff boundary. Microscopically both thermal and fluorescence photons arise from spontaneous emission; an ensemble name alone does not determine the tariff. The subsystem of the act, pointer basis, and physically realized local preparation register must be specified (§ 4.1). The controlled $|+\rangle_{\mathrm{ptr}}$ test in § 6 fixes these data explicitly; for an equilibrium mode their ontic specification remains a condition on the model.

Relation to existing data. There is no published *direct* experimental bound on the tariff by T and the postulate $Q_i = kT\ln(1/p_i)$; below—why adjacent measurements neither exclude nor duplicate the proposed test (§ 6).

Optics. Single-photon calorimetry on TES (Transition Edge Sensor) [30] resolves photon energy at 1550 nm to $\sim$ 0,1–1 eV, but at $T_{\mathrm{bath}} \sim$ 0,1–0,3 K. At tariff "temperature of the act environment" the excess $kT\ln 2$ at such T ($\sim 10^{-5}$ eV per bit) is three to four orders below TES resolution. Cold TES data do *not* bound the tariff by T; the heralded-MZ test (§ 6) with event-triggered calorimetry does not repeat existing single-photon measurements.

Laser radiometry. The primary scale of optical power compares warm thermal detectors with cryogenic electrical-substitution radiometers ($T_{\mathrm{bath}} \sim$ 4–5 K, tariff $kT\ln 2$ there $\sim$ 0,03% of E_γ); agreement of services and detector types is 0.02–1.2%. Naive reading "coherent laser $\Rightarrow$

extra $\sim 2\%$ heat at 300 K" does *not* follow from the model: the tariff pays not field coherence in the amplitude basis but superposition *in the pointer basis of the act* (§ 3)—$|+\rangle_{\text{ptr}}$ vs $I/2_{\text{ptr}}$. An ordinary power meter: one beam, one spot, no 50/50 splitter and no controlled inter-zone phase; this is not the $|+\rangle_{\text{ptr}}$ protocol from § 6 and not a bound on the tariff by T for such a state. The proposed heralded-MZ test (§ 6) is the differential $|+\rangle_{\text{ptr}} - I/2_{\text{ptr}}$ on single-photon qubits; radiometry did not reproduce it.

Bounds on heating of collapse models. Experiments on GRW/CSL/DP [8, 31] (cold atoms, cantilevers, underground setups) bound *any* excess heating from spontaneous localization—"background" heating of the collapse model itself. Heat from Bit asks something else: how much heat does *irreversible fixation of a pointer outcome* give if before the act there was coherence in the pointer basis ($H > 0$). Existing bounds would refute Heat from Bit only if in those setups many pointer bits were destroyed per event. Usually this does not happen: the system is either already a classical mixture ($H \approx 0$), or a "large" superposition is not in the pointer basis where the tariff is counted—then $H = 0$ and additional heating by Heat from Bit is also ≈ 0. Measured zero heating in GRW/CSL/DP tests does not forbid the model at tariff by T. In § 6 H and the number of bits per act are made large intentionally.

Strong globally conditioned variant. If rule (iii) of § 4.1 is adopted instead of the working local rule, Alice's measurement remotely selects for Bob either $\{|0\rangle, |1\rangle\}$ or $\{|+\rangle, |-\rangle\}$ while $\rho_B = I/2$ in both cases. Bob's idealized heat differential then equals the MZ signal (17); with the parameters of § 6, $\tau = (\text{NEP}/\Delta P)^2 \approx 83$ s means only SNR ≈ 1, not one reliably transmitted bit. This would be controllable superluminal signaling and would require a separate global causal-order law, such as a preferred foliation. It is *not* a consequence of nonlinear C_r and is not part of the main model. Bohmian mechanics illustrates preferred foliation with hidden nonlocality but does not itself permit controllable signaling; relativistic rGRWf, by contrast, is constructed without a preferred foliation [32, 33]. These theories are comparisons, not proofs that either structure is necessary.

Falsification. Direct test of the model—§ 6.

Comparison with neighboring approaches. From measurement engines [34, 35, 36, 37, 38, 39] the Heat from Bit model differs by autonomy of the act without apparatus; from quantum thermodynamics of projections [19]—by the postulate of *objective* collapse and tariff in the pointer basis of the act, not a unitary measuring device; from information reservoirs [40, 41, 42, 43]—by duality (bit as energy carrier at $\Delta S_{\text{tot}} = 0$); from GRW/CSL/DP [8]—by organizing heating by bit flux ($kT \ln 2$ per ΔN_{bit}): in standard formulations their heating is diffuse and does not correlate with informational acts (diffuse background by itself does not cancel the Heat from Bit model).

Local first law. The first law is violated only locally, in a sub-τ_{th} act of non-unitary fixation (§ 5); globally the second law is preserved—δQ is dissipated in the reservoir with entropy $k\Delta H$. Per-event energy imbalance is below calorimeter resolution; heralded MZ measures systematic excess over the ensemble (§ 6).

Table 1. Comparison of standard QM, GRW/CSL/DP models, and the Heat from Bit model.

	Standard QM	**GRW/CSL/DP**	**Heat from Bit**
Objective collapse	no	yes	yes
First law	yes	no	locally in act
Second law	yes	yes	yes
Heat in objective act	0	diffuse heating	$\langle Q \rangle = kTH$
Distinctive test	—	Gran Sasso; LISA Pathfinder; ultracold atoms	heralded MZ

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